

Metabolic Responses to Sulfur in Lucinid Bivalves¹

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SYNOPSIS. Lucinid bivalves occur widely in habitats ranging from subtidal to deep ocean basins. The evidence reported to date supports the contention that all lucinids contain intracellular, sulfur-oxidizing bacterial symbionts which contribute substantially to the nutrition of the intact association. These burrowing bivalves are found in both high and low sulfide habitats. Examinations of sulfur compounds in the hemolymph of lucinids reveal that thiosulfate may be a key intermediate in the metabolism of sulfide in all members of this family. The presence of free sulfide in the hemolymph of both freshly collected and experimental clams suggests that the total sulfide oxidation/detoxification capacity of the lucinids may be substantially lower than that found in other sulfur bacteria/invertebrate symbioses. Intracellular granules catalyze the oxidation of sulfide in *L. floridana*, one apparent line of sulfide toxicity defense in this species. These electron-dense granules occur in high densities in the bacteriocytes (bacteria-containing gill cells) of all lucinids. Thiosulfate produced during this detoxification may be available for further oxidation by the abundant symbionts housed in these same cells. Hemolymph thiol and elemental sulfur levels in clams from low sulfide habitats suggest that these animals have evolved mechanisms for sulfur acquisition. In sulfur-free seawater, sulfur-starved *Lucina floridana* are infrequent, periodic ventilators, but in the presence of sulfur show a significant increase in ventilation frequency. This periodic ventilation may be a trait of all lucinids that plays a substantial role in reducing metabolic costs during low nutrient availability.

INTRODUCTION

The bivalve family Lucinidae, of the superfamily Lucinacea, is an ancient, burrow-dwelling group that occurs globally over a wide range of marine habitats. Intracellular bacteria have been observed in the gills of all lucinids examined to date (Felbeck *et al.*, 1981; Cavanaugh, 1983; Berg and Alatalo, 1984; Fisher and Hand, 1984; Dando *et al.*, 1985, 1986a; Dando and Southward, 1986; Schweimanns and Felbeck, 1985). The entire suite of key variables used to assign chemoautotrophic status to this recently discovered type of symbiosis (Fisher, 1990) has not been determined in all lucinids examined. However, the results of EM stud-

ies, assessments of enzymatic capacity for carbon fixation and sulfur metabolism, and/or the determination of stable carbon isotope ratios in the lucinids supports the contention that the symbionts are chemoautotrophic sulfur-oxidizing bacteria (Table 1).

In their natural habitat adult lucinids are always found in association with their symbionts, a condition common to all marine invertebrates with intracellular, sulfur-bacteria symbionts that have been described over the past 15 years. Many authors regard this highly-integrated intracellular symbiosis as an obligate one (Schweimanns and Felbeck, 1985; Reid and Brand, 1986; Southward, 1987; Fisher, 1990), an opinion with which I am in agreement. To my knowledge, Dando *et al.* (1986a) are the only authors to have ever reported the apparent absence of symbionts and related chemoautotrophic enzymatic activity in a lucinid,

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TABLE 1. *Lucinidae with bacterial symbionts.*

Species	EM	³⁵ S	Enzymes	16S-rRNA	S-stimulated carbon fixation	Habitat [Sulfide], μ M	Ref.
<i>Anodontia philippiana</i>	+		Ru	+			10, 14, 20
<i>Codakia costata</i>	+		Ru	+			10, 14, 20
<i>Codakia orbicularis</i>			Ru, ATPS, Rk	+			1, 9, 10
<i>C. (Ctena) orbiculata</i>			Ru				20
<i>Linga (Lucina) pennsylvanica</i>			Ru				1
<i>Loripes lucinalis</i>	+		Ru, APS, ATPS				7, 18
<i>Lucina floridana</i>	+	+	Ru, APS, ATPS	+	+	0.3, 2.5	8, 11, 13, 15, 22
<i>Lucina nasula</i>			Ru				1
<i>Lucina radians</i>			Ru				20
<i>Lucinella divaricata</i>	+		Ru, APS, ATPS				7, 17
<i>Lucinoma aequizonata</i> ^a	+	+	Ru, APS, ATPS	+		0-8	3, 4, 8, 9, 10, 12, 22
<i>Lucinoma annulata</i>	+			+			8, 9, 10
<i>Lucinoma atlantis</i> ^b							16
<i>Lucinoma borealis</i>	+	+	Ru, APS, Rk		+	0.2, <1	6, 21
<i>Myrtea spinifera</i>	+	+	Ru, APS, ATPS		+	<1	5, 21
<i>Parvilucina multilineata</i>	+		Ru				14, 20
<i>Parvilucina tenuisculpta</i>	+	+	Ru, ATPS				12, 19
<i>Pseudomiltha</i> sp.	+	+	Ru, APS, ATPS				2

Note: Abbreviations: EM: bacterial symbionts viewed via electron microscopy; ³⁵S: substantial elemental sulfur stores in gills; Enzymes: APS = APS reductase, ATPS = ATP Sulfurylase, Rk = Ribulose 5'kinase, Ru = Ribulose biphosphate carboxylase/oxygenase; 16S rRNA = bacterial rRNA extracted from intracellular symbionts; Habitat [sulfide], μ M: interstitial sulfide levels in bivalve habitat(s). (Acid-labile sulfides not included.) Note: sulfide levels determined by different methods may not be comparable. Habitat sulfide varies with temperature, season, sediment depth, microbial population, oxygen availability, salinity, sediment organics.

^a Identified incorrectly as *L. annulata* in references 3, 9, 19.

^b Only stable carbon isotope (¹²C:¹³C) ratios available for this sp.

References: (1) Berg and Alatalo, 1984; (2) Brooks *et al.*, 1987; (3) Cary *et al.*, 1989; (4) Cavanaugh, 1983; (5) Dando *et al.*, 1985; (6) Dando *et al.*, 1986a; (7) Diouris *et al.*, 1988; (8) Distel and Felbeck, 1987; (9) Distel *et al.*, 1988; (10) Distel *et al.*, 1993; (11) D. L. Distel, unpublished data; (12) Felbeck *et al.*, 1981; (13) Fisher and Hand, 1984; (14) Giere, 1985; (15) Hand, 1987; (16) Kennicutt *et al.*, 1985; (17) LePennec *et al.*, 1987; (18) LePennec *et al.*, 1988; (19) Reid and Brand, 1986; (20) Schweimanns and Felbeck, 1985; (21) Spiro *et al.*, 1986; (22) Vetter, 1985.

this after 12 months starvation of *Lucinoma borealis* maintained in sulfur-free seawater. According to the hypothesis of Reid and Brand (1986), in two related groups in the same superfamily, the Ungulinidae and Thyasiridae, the symbiotic condition has been either lost or substantially regressed. However, well-developed chemoautotrophic symbioses do exist in the Thyasirids as well (Dando and Southward, 1986; Southward, 1986; see Fisher, 1990, for review).

MORPHOLOGY AND BEHAVIOR

All lucinids show a substantial reduction of the labial palps and alimentary tract (Allen, 1958), though they are not gutless like numerous other well-studied invertebrate/sulfur bacteria symbioses (reviewed

in Fisher, 1990). These bivalves are also characterized by large, fleshy, granule-containing gills. The gill bacteriocytes (symbiont-containing cells) are arranged in cylindrical channels perpendicular to the long axis of the gill filaments, an arrangement which confers a large surface to volume ratio to this organ (Distel and Felbeck, 1987). The capacity for carbon fixation has been shown enzymatically in all lucinid gills examined for this trait to date (see Table 1). This capacity, and negative stable carbon isotope ratios (Fisher, 1990), suggest that the hosts are provided a significant proportion of their reduced carbon needs via translocation or symbiont digestion. Cary *et al.* (1989), based upon $\delta^{13}\text{C}$ values of isolated symbionts and host tissue from *Lucinoma aequizonata*, calculated that the symbionts

account for about 75% of the nutrition of the association, with exogenous carbon accounting for the remaining 25%. Further discussion of the $\delta^{13}\text{C}$ ratios and other aspects of carbon metabolism in the lucinids is presented in an excellent review by Fisher (1990) and will not be addressed in this paper.

Allen (1953, 1958) has described the burrowing behavior and detailed anatomy of the lucinids. In place of an inhalant siphon (absent in this group), the foot is used to penetrate the overlying sediments to form a mucus-lined anterior inhalant 'tube' for access to the overlying seawater (Fig. 1). Ventral to the posterior exhalant siphon lies an unfused portion of the mantle, the incurrent aperture (termed the "posterior inhalant aperture" by Britton, 1970), through which pseudofeces are eliminated and pore water from the sediments could diffuse. These unusual morphological traits are attributed by Reid and Brand (1986) to be associated with their symbiotic condition. Dando *et al.* (1985) note that a new inhalant tube is formed by *M. spinifera* every few days. The lucinid foot can also be used to probe up to 3 body lengths deep into the substrate below the burrow: Cary *et al.* (1989) report that *L. aequizonata* dug 4–5 holes per week in artificial habitats constructed in their lab. In low sulfide habitats, this activity is potentially extremely important for sulfide acquisition, and will be further discussed. Figure 1 shows a lucinid positioned in its burrow.

SEDIMENT CHEMISTRY IN LUCINID HABITATS

The lucinids occur in an extremely wide range of habitats, from highly reducing, sulfide-rich muds in shallow, warm eelgrass beds (Berg and Alatalo, 1984; Fisher and Hand, 1984) to deep marine basins or subtidal zones where sulfide is extremely low or undetectable (Britton, 1970; Dando *et al.*, 1985, 1986a; Cary *et al.*, 1989). Though general descriptions of these habitats are common, few authors have reported in detail the critical chemical characteristics (levels of sediment and porewater oxygen, sulfide, and other partially oxidized sulfur species) of the lucinid habitats.

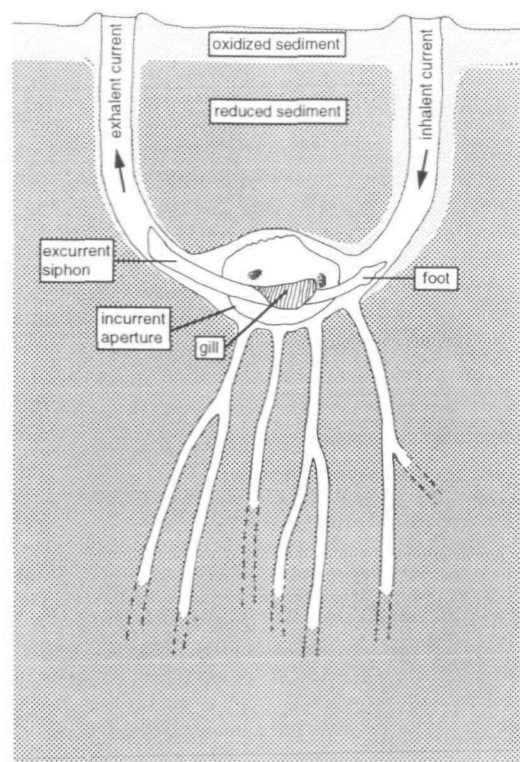


FIG. 1. Diagram of the position of a "typical" lucinid in its burrow, drawn from the x-rays of Stanley (1970). Anterior inhalant tube and tunnels beneath burrow are excavated by the greatly extensible foot. (Drawing by R. E. Kochevar.) (Reprinted with permission from Fisher, 1990.)

High sulfide habitats

One lucinid habitat reported to have very high levels of porewater sulfide ($> 25 \mu\text{M}$) occurs off the southern coast of the Florida panhandle in shallow, warm waters in highly productive, detritus-rich seagrass beds (Kikuchi and Peres, 1977). There, high densities of *Lucina floridana* (83 ± 11 per m^2 ; Fisher and Hand, 1984) live in burrows tightly wedged among the roots of *Thalassia* and *Ruppia* (Britton, 1970). Sulfide levels of 1.67 and 2.49 mM, respectively, were measured in these habitats (Fisher and Hand, 1984). Though Britton (1970), and Fisher and Hand (1984) report sparse *L. floridana* densities in regions with sparse vegetation, D. Kraus and coworkers find high densities of these clams at depths of 40–60 cm in seagrass-free sediments (per-

sonal communication). This difference is likely explained by differences in the depth of collection. Thus, while this distribution difference does not support seagrass roots as a habitat requirement, the oxygen generated by these plants (Teal and Kanwisher, 1966; Armstrong, 1970) may play an important role in both detoxification and oxidation of the respiratory toxin sulfide for sulfur-based energy metabolism for *L. floridana*, as suggested by Fisher and Hand (1984). Berg and Alatalo (1984) report that *Codakia orbicularis* also live in "high sulfide" habitats (actual sulfide levels not reported) in shallow subtidal seagrass beds in the Bahamas.

Low sulfide habitats

In contrast, the habitats of other lucinid species are reported to have extremely low levels ($10\ \mu\text{M}$ or $<$) of dissolved porewater sulfur compounds. (Dando *et al.*, 1985, 1986a, b; Cary *et al.*, 1989), raising the question of how these symbiotic species obtain sufficient sulfur to meet their chemoautotrophic requirements. Dando *et al.* (1985, 1986a, b) report on two lucinids, *Myrtea spinifera* and *Lucinoma borealis*, which reside in a low sulfide habitat in a fjord in Norway. *M. spinifera* occur in high densities (up to 25 per m^2) in muddy sand at depths of about 33 m, while *L. borealis* (densities vary from 2 to 110 per m^2 depending on location) burrow in sand low in the intertidal zone in or adjacent to *Zostera* beds. Free sulfide was extremely low, and ranged from $0.0\text{--}0.08\ \mu\text{moles } 1,000\ \text{cm}^{-3}$, and thiosulfate from $0\text{--}0.12\ \mu\text{moles } 1,000\ \text{cm}^{-3}$, in the sediments, while acid-labile sulfide ($150\text{--}14,000\ \mu\text{moles } 1,000\ \text{cm}^{-3}$) and elemental sulfur ($80\text{--}4,000\ \mu\text{moles } 1,000\ \text{cm}^{-3}$) were orders of magnitude higher. At burrow depths for *L. borealis* (15–20 cm), acid-labile sulfide predominated among the sulfur compounds tested. Despite the low levels of dissolved sulfide, the occurrence of elemental sulfur in the gills of these clams demonstrates sulfide availability and the active metabolism of this compound.

Another lucinid habitat depauperate in dissolved sulfide is described by Cary *et al.* (1989). These authors studied the chemical characteristics of the deep basin habitat

(490–510 m) in which *Lucinoma aequizonata* resides in the Santa Barbara basin. This region is severely oxygen limited, with an average concentration of less than $20\ \mu\text{M O}_2$. Emery and Hulseman (1962) also report lucinids inhabiting low oxygen (approximately $5\text{--}24\ \mu\text{M}$) habitats. While other authors have reported the sulfur, oxygen and sediment profiles of numerous lucinid habitats (Dando *et al.*, 1985, 1986a), Cary *et al.* (1989) present not only an extensive habitat profile, but also the physiological and metabolic responses of *L. aequizonata* to sulfur. Their findings will be reviewed in detail below.

SULFUR METABOLISM IN THE LUCINIDS

There are very few studies describing the physiological responses to sulfur compounds in the lucinids. The majority of the biochemical studies on this group have focused on identifying and quantifying the key enzymes of carbon autotrophy (Ribulose biphosphate carboxylase/oxygenase and Ribulose 5' kinase) and bacterial sulfur metabolism (ATP sulfurylase, APS reductase) (Table 1; Fisher, 1990). Most of the sulfur-focused studies have examined elemental sulfur stores, and/or the degree of stimulation of carbon fixation in the symbiont-containing gills by various sulfur compounds. Short-term physiological studies using intact clams are difficult to conduct due to this group's notorious reputation for "clamping up" for many hours or even days at a time. These animals are impressive anaerobes, and easily survive 48 hours of anaerobiosis (personal observation). To my knowledge no specific biochemical or physiological studies have been done on the anaerobic poise of these animals. Given the habitat characteristics, burrow depths and relative immobility of these animals, detailed studies that focus on the mechanisms of sulfur and energy metabolism of the host and symbionts would be a valuable addition to this field. Discussion of alternative electron acceptors during sulfur oxidation are addressed later in this review.

Myrtea spinifera

Dando *et al.* (1985) measured sulfide uptake in excised gills of both the symbiotic

clam *M. spinifera* and the non-symbiotic *Arctica islandica* using ^{35}S -sulfide. They report uptake rates in *M. spinifera* ranging from 0.39–1.15 $\mu\text{moles g}^{-1}$ wet wt hr^{-1} over a 3 hr incubation. These rates are about one tenth those measured for the very active protobranch *Solemya reidi* (3.07 $\mu\text{mol/g}^{-1}$ hr^{-1} for whole clams, of which approximately 25% of the wet weight is gill; Anderson *et al.*, 1987), which lives in a very high sulfide habitat (up to 22 mM; Childress and Lowell, 1982). There was a much greater proportion of metabolized sulfur intermediates in *M. spinifera* gills (ave. 81%) than *A. islandica* gills (ave. 44%), and these authors (Dando *et al.*, 1985) concluded that, though sulfide was taken up by both species, only the symbiotic clam is able to metabolize a great proportion of the sulfide to other sulfur products.

Lucinoma aequizonata

Cary *et al.* (1989) analyzed the habitat chemistry and hemolymph sulfur compounds in *L. aequizonata* from the Santa Barbara basin. Despite barely detectable levels of dissolved reduced sulfur compounds ($<10 \mu\text{M}$) in the interstitial waters, freshly collected animals had elemental sulfur in their gills, and thiosulfate and sulfide in the circulating hemolymph. While hemolymph sulfide was approximately equivalent to the low habitat levels, they found thiosulfate at concentrations of up to 62 μM in the hemolymph and 78 μM in the mantle fluid.

These authors also studied the sulfide tolerance and growth rates of *L. aequizonata* maintained in four different levels of sulfide, and the metabolic sulfur products resulting from incubations under each condition. After four weeks in seawater (SW), or SW with 10, 30 or 50 μM added sulfide, clams at all levels of sulfide had substantial levels of elemental sulfur (S^0) in their gills (up to 3% D.W.), whereas S^0 was undetectable after about 3 weeks in animals maintained in sulfide-free SW. Clams incubated in 50 μM sulfide had the highest mortality and no measurable growth (those in sulfide-free SW also showed no growth). Ten μM sulfide supported the highest growth rates, and no

mortality was seen in this group. The oxygen maintained in the burrows was greater than that of the sediments (0 μM), but less than that of the overlying water (20 μM). Juveniles held in air-saturated, sulfide-free SW died within 5 days, whereas those held in SW with 20 μM oxygen all survived (S. C. Cary, personal communication). In contrast, *L. floridana* can survive for months in air-saturated SW (A. Anderson, unpublished data). Thus, compared to the high sulfide tolerance and sulfide "optimum" (100 μM) in the protobranch bivalve *Solemya reidi* (Anderson *et al.*, 1987), the sulfide "optimum" of *L. aequizonata* is in the low micromolar range. In comparing these two species, it is apparent that their very different sulfide tolerance levels have evolved in response to the chemistry of the habitats in which they reside.

Cary *et al.* (1989) also assayed habitat porewater for electron acceptors other than oxygen available for sulfide oxidation. As stated earlier, oxygen levels in the overlying waters of this basin are very low (20 μM). They found no significant sulfate reduction in the anoxic sediments and no clear sulfide boundary. Lack of these features was attributed to substantial bioturbation. Levels of nitrate, iron, and manganese, all potential alternate electron acceptors, were very low in the sediments. Metal sulfides were also present, indicating that sulfate reduction does indeed take place, even if at low levels. The authors suggest that high levels of sediment iron may be responsible for removing the sulfide by precipitation as it is generated, a sulfur source which could be later remobilized. Finally, they suggest that nitrate in the overlying SW (40 μM) may serve as an electron sink for sulfide oxidation in *L. aequizonata*, which has high levels of nitrate reductase in its tissues (J. Stein, unpublished data), much as occurs in some free-living bacteria (Schedel and Truper, 1980).

A recent paper by Hentschel *et al.* (1993) shows that nitrate removal and nitrite appearance occur in anaerobic, 50 μM nitrate incubations of isolated *L. aequizonata* symbionts. The importance of nitrate respiration in aerobic, 1 mM nitrate incubations in this same study is difficult to assess because of the unrealistically high nitrate

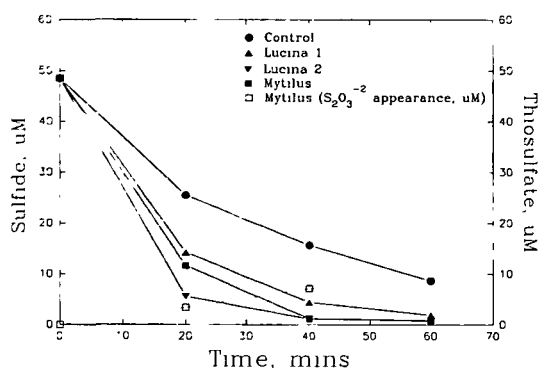


FIG. 2. Sulfide oxidation in excised bivalve gills. Gills of *L. floridana* ($n = 2$) and *M. californianus* ($n = 1$) (non-symbiotic bivalve) were incubated in 50 μM sulfide. Separate flasks of *L. floridana* gills ($n = 2$) and *M. californianus* gill ($n = 1$) were incubated at 20°C in a buffered (20 mM Hepes, pH 8.0), 0.45 μm -filtered seawater solution of 40 μM sulfide. Each incubation medium was sampled every 20 minutes for the one hour. A control flask (no gills) was used to monitor spontaneous sulfide oxidation rates. Samples were derivatized with the monobromobimane and the sulfur compounds identified and quantified via HPLC as described in Vetter *et al.* (1989). Points represent thiol (sulfide or thiosulfate) levels in the incubation medium at time = X. No thiosulfate appeared in the control or lucinid incubations over the experimental period.

concentration. Hentschel *et al.* (1993) also report that symbiont respiration was not stimulated by the addition of low concentrations of sulfide or thiosulfate, raising the question of the utilization of nitrate for sulfur oxidation. In contrast, in calorimetric studies with intact *L. floridana*, we found no increase in metabolic heat dissipation in anoxic, 500 μM thiosulfate incubations containing either 50 or 500 μM nitrate (Anderson and Hand, in preparation). Thus, while it has been shown that many of the marine invertebrates containing symbionts have the enzymatic capacity to reduce nitrate (Felbeck *et al.*, 1981), and that the nitrate reductase in *L. aequizonata* is a dissimilatory enzyme (Hentschel *et al.*, 1993), more studies demonstrating dissimilatory nitrate reduction at environmentally-realistic levels of oxygen and nitrate will need to be done to demonstrate that these pathways are utilized. If it could be shown that the host tissues in *L. aequizonata* provide an anaerobic environment for their symbionts, the data of Hentschel *et al.* (1993) would

strongly support their hypothesis that nitrate respiration is important in this species.

Lucina floridana

Recent studies in our laboratory demonstrate the oxidation of sulfide by excised gills of *L. floridana* (Fig. 2). We used gills of *Mytilus californianus* as a non-symbiotic control. *L. floridana* were collected in St. Joseph's Bay in Southern Florida and held in sulfur-enriched sediments (Anderson *et al.*, 1987) at room temperature in aquaria in our laboratory. *M. californianus* were collected in Newport, Oreg. and held in flowing SW trays at 15°C. We derivatized and quantified thiols using the monobromobimane HPLC technique (Vetter *et al.*, 1989).

L. floridana and *M. californianus* both oxidize sulfide at a rate greater than background spontaneous oxidation (Fig. 2). In *L. floridana* gill incubations, sulfide levels declined during the first 60 min, and no partially-oxidized sulfur products appeared in the medium (Fig. 2). However, although sulfide also declined during incubations of *M. californianus* gills, both sulfite and thiosulfate appeared in increasing concentrations over the first hour of incubation, after which nearly all the sulfide had disappeared. Sulfide oxidation capacity in *M. californianus* gills was not a surprising finding, as many compounds such as hemes, proteins, etc., are known to catalyze sulfide oxidation (see Somero *et al.*, 1989, for review). These results show that in *L. floridana*, the products of sulfide oxidation in the tissues are further oxidized by the symbionts, while in *M. californianus* both sulfite and thiosulfate are excreted into the medium. In a separate experiment, we showed that excised gills from *L. floridana* also take up thiosulfate (Anderson and Hand, in preparation), a compound important in the sulfide oxidation pathway, as suggested by the studies described above.

We also incubated whole clams in flowing SW (buffered with 20 mM Hepes, pH 8.0) with 40 μM sulfide, and assayed the hemolymph for thiols resulting from sulfide oxidation using monobromobimane (see above). We found both sulfide and thiosulfate in the hemolymph of these animals (Fig. 3), as did Cary *et al.* (1989) in *L. aequizon-*

ata. Although hemolymph sulfide concentrations in all animals were below sulfide exposure levels (40 μM), thiosulfate increased substantially over the 5 hr incubation to 380 μM . Even though Anderson *et al.* (1987) also found high levels of thiosulfate in the hemolymph of *S. reidi* incubated in 100 μM (oxic) sulfide, sulfide did not appear in the hemolymph of these clams, and the animal tissues remained aerobic (Anderson *et al.*, 1989). These differences suggest a higher sulfide oxidation capacity in *S. reidi* than *L. floridana*, and that anaerobic metabolism may play a critical role in the lucinids during sulfide exposure.

Sulfide detoxification

Only one study has addressed a possible mechanism for sulfide detoxification in the lucinids (Anderson and Hand, 1990; see below), and this remains an area for future studies. J. J. Childress (unpublished data) found no sulfide binding in hemolymph of *L. aequizonata* (as is found in other invertebrate/sulfur bacteria symbioses; see Somero *et al.*, 1989, for review). The phenomenon of sulfide binding by the intracellular hemoglobins of the lucinids is addressed in an accompanying paper by D. Kraus in this symposium. As mentioned above, sulfide oxidation is catalyzed by a wide variety of biological compounds and occurs in many animals that do not harbor intracellular sulfur bacteria symbionts (Vetter *et al.*, 1987; Bagarinao and Vetter, 1990). That isolated mitochondria from both marine invertebrate (symbiotic and non-symbiotic) species and fishes have shown some sulfide oxidation capacity (Powell and Somero, 1986; Bagarinao and Vetter, 1990; O'Brien and Vetter, 1990; O'Brien, 1991) suggests that this might be a common mechanism for detoxification in many eucaryotes. Mitochondrial sulfide oxidation has not yet been investigated for the lucinids.

Like all lucinids, *L. floridana* harbor numerous pigmented granules in their gills (Allen, 1958). These are found predominantly in the bacteriocytes (Fisher and Hand, 1984; Hand and Anderson, 1993), but are not exclusive to those cells (personal observation). Granules from *L. floridana* contain proteins, heme, and numerous minerals (sil-

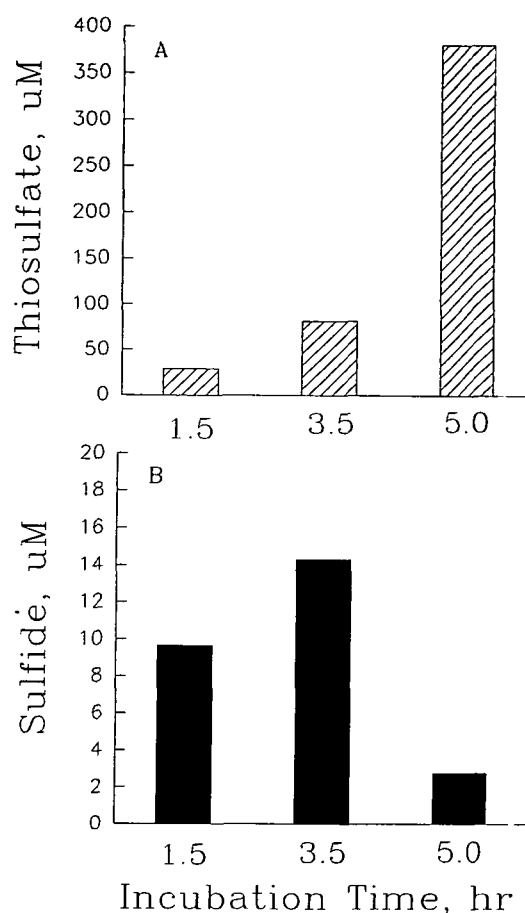


FIG. 3. Blood thiols in sulfide-incubated *L. floridana*. A. Blood thiosulfate. B. Blood sulfide. Each bar represents the thiols in the blood of one individual clam after x hours in flowing 40 μM sulfide.

ica, feldspar, pyrite, calcium phosphate, hornblende, and others), some of which are found in the habitat sediments (Anderson and Hand, 1990; Anderson and Hand, in preparation). Distel and Felbeck (1987) note the resemblance of these granules to residual lysosomal bodies because of the associated membranous whorls when viewed by EM. Preliminary studies using the artificial electron acceptor benzyl viologen showed that isolated granules from *L. floridana* catalyzed the oxidation of sulfide (Anderson and Hand, 1990). Recent studies of sulfide oxidation by pigment granules using monobromobimane and HPLC analysis confirm these results, and reveal thiosulfate as one

of the products of this granule-mediated oxidation (Anderson and Hand, in preparation). Powell and Arp (1989) showed that nonhemoglobin hematin granules from the marine echiuran *Urechis caupo* also stimulated the oxidation of sulfide, as do the granules from the symbiont-containing clam *S. reidi* (O'Brien, 1991). Allen (1958) proposed that these lucinid granules are primarily waste products. Fisher and Hand (1984) suggested that these granules might serve an oxygen delivery role in the bacteriocytes of *L. floridana*, and Giere (1985) hypothesizes that they are lysosomal products and serve as a sink for metal detoxification. Whatever their primary role and fate, these granules may play a secondary function in the detoxification and metabolism of sulfide in this bivalve family.

Respiratory responses to sulfur in L. floridana

L. floridana exhibits a pattern of periodic ventilation (Anderson and Hand, in preparation). This ventilatory pattern may be a characteristic of all lucinids, as it has also been observed in *L. aequizonata* (S. C. Cary, unpublished data). In the presence of thiosulfate, we found that sulfur-starved *L. floridana* markedly increased their ventilatory frequency (Anderson and Hand, in preparation). Studies in progress will examine whether this response is specific to sulfur compounds, or whether it is a response to any type of nutritive substance, for example algae or free amino acids, present in the incurrent water. This pattern of infrequent ventilations in the absence of available "fuel" (sulfur compounds) serves as an energy conserving strategy in *L. floridana*. It is conceivable that periodic ventilation in the lucinids might also be adaptive in avoiding sulfide poisoning, as their detoxification capacity is clearly insufficient to oxidize entering sulfide even at low external concentrations (see above).

STRATEGIES FOR ACQUISITION OF SULFIDE IN LOW SULFIDE HABITATS

Dando *et al.* (1985) calculated that the low levels of dissolved sulfur compounds in the interstitial waters where *M. spinifera* and *L. borealis* reside are insufficient, at biolog-

ically reasonable interstitial water pumping rates, to support chemoautotrophic nutrition in these species. These authors suggest that sediment-bound sulfides consisting of finely divided ferrous sulfide and pyrites that surround the inhalant tube may become partially oxidized via incurrent seawater and microbial activity, and products such as thiosulfate and sulfite made available in the incurrent respiratory stream. Such sediment bound sulfides have been shown to dissolve in oxygenated SW at neutral to slightly acidic pH (Nelson *et al.*, 1977; Goldhaber, 1983). An additional sulfur acquisition mode suggested by Dando *et al.* (1985, 1986a) is through the incurrent aperture: dissolved interstitial sulfur compounds, potentially augmented by the degradation of pseudofeces including fine sediment particulates in the burrow, could be taken up through this aperture, while oxygen is accessed through the inhalant tube. Distel and Felbeck (1987) point out the possible benefits of the ctenidial morphology, where oxygenated water must first pass the ciliated epithelial layer and intercalary cells before reaching the bacteriocytes. This arrangement might protect sensitive bacterial enzymes from oxygen toxicity, as might the presence of gill hemoglobin (Dando *et al.*, 1985; Kraus and Wittenberg, 1990). Distel and Felbeck (1987) also contend that this morphological arrangement implies that sulfide detoxification must occur in the epithelial tissues, and that the protected bacteriocytes probably have no role in sulfide detoxification. Though we now know that bacteriocyte granules do oxidize sulfide (Anderson and Hand, in preparation), specific mechanisms of toxicity defense in the epithelial tissues (mitochondrial, enzymatic) have not yet been examined in the lucinids.

Cary *et al.* (1989) reject the sulfur acquisition strategies of Dando *et al.* (1985) for *L. aequizonata*, as the metal sulfides in sediments in the Santa Barbara basin where these clams live were below detection limits, probably due to extensive bioturbation in this habitat. However, like the lucinids studied by Dando *et al.* (1985, 1986a), *L. aequizonata* was found to have elemental sulfur in its gills, and sulfide and thiosulfate in its hemolymph, indicating access to envi-

ronmental sulfide despite the extremely low levels in the interstitial water. These authors suggest that *L. aequizonata* may take advantage of randomly dispersed "pockets" of sulfurous mud in the sediments, accessing these via their probing feet, an hypothesis suggested for the lucinids by Turner (1985). This hypothesis is certainly supported by the evidence presented by Cary *et al.* (1989), and may also be an important route of acquisition for *L. floridana* based on their behavior in laboratory aquaria (A. Anderson, personal observation).

Reid and Brand (1986) offer two additional mechanisms for sulfur acquisition in these sulfide-poor habitats. They report a posterior fusion of the gills with the mantle edge in the lucinid *Parvilucina tenuisculpta*, a morphology they suspect is universal in this bivalve family. This morphological arrangement could cause a bellows-like movement in response to mantle contraction, potentially bringing sulfur-laden water in the deeper sediments into the suprabranchial chamber via the exhalant siphon or incurrent aperture. However, this is not likely to be important in habitats that have been shown to be depauperate in sulfide. A second mechanism suggested is that the low pH in the gut could cause the release of free sulfide from ingested metal sulfides in the sediments, which could then diffuse into the tissues.

Whatever mechanism, or combination of mechanisms, are used to acquire sulfide in these low-sulfur habitats, the access to sulfur is apparent given the data reported. These results strongly support two postulates: not only does active sulfur metabolism occur in these sulfur-poor environs, but sufficient sulfur is acquired to both fulfill immediate metabolic needs and provide elemental sulfur for storage, a situation that does not suggest sulfur starvation.

CONCLUSIONS

The lucinids comprise a large bivalve family that resides in benthic habitats that vary widely in sediment chemistry. It is likely that these symbiotic bivalves have evolved numerous mechanisms for sulfur acquisition in the sulfide-depauperate habitats, as stable carbon isotope data support

a high degree of nutritional reliance on carbon fixed by their symbionts. Detoxification strategies in this family have only begun to be explored; the catalysis of sulfide oxidation by dense aggregates of intracellular granules in *L. floridana* is likely to be a common mechanism in the lucinids. Additionally, the production of thiosulfate during this detoxification provides this energy-rich substrate for further oxidation by the symbionts. Thiosulfate in the hemolymph of freshly collected and experimental animals supports the hypothesis that this compound is also the result of other as yet undefined detoxification strategies and plays an important role in the metabolism of this compound in the lucinids. Finally, infrequent periodic ventilation in the absence of sulfur serves as an effective energy-conserving strategy, especially important in low sulfur habitats, and as a mechanism for sulfide toxicity avoidance in high sulfur habitat species. Increases in ventilation frequency in the presence of sulfur suggest that lucinids may be "opportunistic feeders," *i.e.*, responding to nutrients in the water column. Whether this ventilatory response is specific to sulfur or occurs in the presence of other foods (algae, amino acids) is currently being explored.

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REFERENCES

- Allen, J. A. 1953. Function of the foot in the Lucinacea. *Nature* 171:1117-1118.
- Allen, J. A. 1958. On the basic form and adaptations in the Lucinacea (Eulamellibranchia). *Phil. Trans. R. Soc. Lond. B* 241:421-484.
- Anderson, A. E., J. J. Childress, and J. A. Favuzzi. 1987. Net uptake of CO₂ driven by sulphide and thiosulphate oxidation in the bacterial symbiont-containing clam *Solemya reidi*. *J. Exp. Biol.* 131: 1-32.

- Anderson, A. E., H. Felbeck, and J. J. Childress. 1989. Aerobic metabolism is maintained in animal tissue during rapid sulfide oxidation in the symbiont-containing clam *Solemya reidi*. *J. Exp. Zool.* 256: 130–134.
- Anderson, A. E. and S. C. Hand. 1990. Intracellular pigment granules in the gills of *Lucina floridana* catalyze the oxidation of sulfide. *The Physiologist*, A79.12.
- Armstrong, W. 1970. Rhizosphere oxidation in rice and other species: A mathematical model based on the oxygen flux component. *Physiol. Plant* 23: 623–630.
- Bagarinao, T. and R. D. Vetter. 1990. Oxidative detoxification of sulfide by mitochondria of the California killifish *Fundulus parvipinnis* and the speckled sanddab *Citharichthys stigmaeus*. *J. Comp. Phys. B* 160:519–527.
- Berg, C. J. and P. Alatalo. 1984. Potential of chemosynthesis in molluscan mariculture. *Aquaculture* 39:165–179.
- Britton, J. C., Jr. 1970. The Lucinidae (Mollusca: Bivalvia) of the Western Atlantic Ocean. Ph.D. Diss., The George Washington University.
- Brooks, J. M., M. C. Kennicutt, II, C. R. Fisher, S. A. Macko, K. Cole, J. J. Childress, R. R. Bidigare, and R. D. Vetter. 1987. Deep-sea hydrocarbon seep communities: Evidence for energy and nutritional carbon sources. *Science* 238:1138–1142.
- Cary, S. C., R. D. Vetter, and H. Felbeck. 1989. Habitat characterization and nutritional strategies of the endosymbiont-bearing bivalve *Lucinoma aequizonata*. *Mar. Ecol. Prog. Ser.* 55:31–45.
- Cavanaugh, C. M. 1983. Symbiotic chemoautotrophic bacteria in marine invertebrates from sulfide-rich habitats. *Nature* 302:58–61.
- Childress, J. J. and W. Lowell. 1982. The abundance of a sulfide-oxidizing symbiosis (the clam *Solemya reidi*) in relation to interstitial water chemistry. *Amer. Zool.* 63:A45.
- Dando, P. R., A. J. Southward, E. C. Southward, N. B. Terwilliger, and R. C. Terwilliger. 1985. Sulfur-oxidizing bacteria and haemoglobin in gills of the bivalve mollusc *Myrtea spinifera*. *Mar. Ecol. Prog. Ser.* 23:85–98.
- Dando, P. R. and A. J. Southward. 1986. Chemoautotrophy in bivalve molluscs of the genus *Thyasira*. *J. Mar. Biol. Ass. U.K.* 66:915–929.
- Dando, P. R., A. J. Southward, and E. C. Southward. 1986a. Chemoautotrophic symbionts in the gills of the bivalve mollusc *Lucinoma borealis* and the sediment chemistry of its habitat. *Proc. R. Soc. B.* 227:227–247.
- Dando, P. R., A. J. Southward, E. C. Southward, and R. L. Barrett. 1986b. Possible energy sources for chemoautotrophic prokaryotes symbiotic with invertebrates from a Norwegian fjord. *Ophelia* 26: 135–150.
- Dioris, M., D. Moraga, M. Le Pennec, A. Herry, and A. Donval. 1988. Chimioautotrophie et nutrition chez les Lucinacea, bivalves littoraux de milieu reducteurs. I. Activités enzymatiques des bactéries chimioautotrophes associées aux branchies. *Haliotis* 18:195–205.
- Distel, D. L. and H. Felbeck. 1987. Endosymbiosis in the lucinid clams, *Lucinoma aequizonata*, *Lucinoma annulata* and *Lucina floridana*: A reexamination of the functional morphology of the gills as bacteria-bearing organs. *Mar. Biol.* 96:79–86.
- Distel, D. L., H. Felbeck, and C. M. Cavanaugh. 1993. Evidence for phylogenetic congruence among sulfur-oxidizing chemoautotrophic bacterial endosymbionts and their bivalve hosts. *J. Mol. Evol.* (In press)
- Distel, D. L., D. J. Lane, G. L. Olsen, S. J. Giovannoni, B. Pace, N. R. Pace, D. A. Stahl, and H. Felbeck. 1988. Sulfur-oxidizing bacterial endosymbionts: Analysis of phylogeny and specificity by 16S rRNA sequences. *J. Bacteriol.* 107:2506–2510.
- Emery, K. O. and J. Hulsemann. 1962. The relationship of sediments, life, and water in a marine basin. *Deep-Sea Res.* 8:165–180.
- Felbeck, H., J. J. Childress, and G. N. Somero. 1981. Calvin-Benson cycle and sulphide oxidation enzymes in animals from sulphide-rich habitats. *Nature* 293:291–293.
- Fisher, C. R. 1990. Chemoautotrophic and methanotrophic symbioses in marine invertebrates. *Crit. Rev. Aquatic Sci.* 2:399–436.
- Fisher, M. R. and S. C. Hand. 1984. Chemoautotrophic symbionts in the bivalve *Lucina floridana* from seagrass beds. *Biol. Bull.* 167:445–459.
- Giere, O. 1985. Structure and position of bacterial endosymbionts in the gill filaments of Lucinidae from Bermuda (Mollusca, bivalvia). *Zoomorphology* 105:296–301.
- Goldhaber, M. B. 1983. Experimental study of metastable sulphur oxyanion formation during pyrite oxidation at pH 6.9 and 30°C. *Am. J. Sci.* 283: 193–217.
- Hand, S. C. 1987. Trophosome ultrastructure and the characterization of isolated bacteriocytes from invertebrate-sulfur bacteria symbioses. *Biol. Bull.* 173:260–274.
- Hand, S. C. and A. E. Anderson. 1993. Isolation and characterization of bacteriocytes from a bivalve-sulfur bacterium symbiosis. In P. Kemp, B. Sherr, E. Sherr, and J. Cole (eds.), *Handbook of methods in aquatic microbial ecology*, pp. 33–39. Lewis Publishers, Chelsea, MI.
- Hentschel, U., S. C. Cary, and H. Felbeck. 1993. Nitrate respiration in chemoautotrophic symbionts of the bivalve *Lucinoma aequizonata*. *Mar. Ecol. Prog. Ser.* 94:35–41.
- Kennicutt, M. C., II, J. M. Brooks, R. R. Bidigare, R. R. Fay, T. L. Wade, and T. J. McDonald. 1985. Vent type taxa in a hydrocarbon seep region on the Louisiana slope. *Nature* 317:351–353.
- Kikuchi, T. and J. M. Peres. 1977. Consumer ecology of seagrass beds. In C. P. McElroy, C. Helferich (eds.), *Seagrass ecosystems: A scientific perspective*, pp. 107–134. Marcel Dekker, Inc., New York.
- Kraus, D. W. and J. B. Wittenberg. 1990. Hemoglobins of the *Lucina pectinata*/bacteria symbiosis. *J. Biol. Chem.* 265:16043–16053.
- Le Pennec, M., A. Herry, M. Diouris, D. Moraga, and A. Donval. 1987. Chimioautotrophie bactérienne chez le mollusque bivalve littoral *Lucinella divaricata* (Linné). *C. R. Acad. Sci. Paris Sée III* 305:1–5.

- Le Pennec, M., A. Herry, M. Diouris, D. Moraga, and A. Donval. 1988. Chimioautotrophie et nutrition chez les *Lucinacea*, bivalves littoraux de milieux reducteurs. II. Caracteristiques morphologiques des bacteries symbiotiques et modifications structurales adaptatives des branchies de l'hote. *Haliotis* 18:207-217.
- Nelson, M. B., J. A. Davis, M. M. Benjamin, and J. O. Leckie. 1977. The role of iron sulfides in controlling trace metals in anaerobic sediments: Oxidative dissolution of ferrous monosulfides and the behavior of associated trace metals. Department of Civil Engineering, Stanford University, Stanford.
- O'Brien, J. 1991. Adaptations of mitochondria to environmental stresses: Sulfide detoxification and heat resistance. Ph.D. Diss., University of California, San Diego.
- O'Brien, J. and R. D. Vetter. 1990. Production of thiosulfate during sulphide oxidation by mitochondria of the symbiont-containing bivalve *Solemya reidi*. *J. Exp. Biol.* 149:133-148.
- Powell, M. A. and A. J. Arp. 1989. Hydrogen sulfide oxidation by abundant non-hemoglobin heme compounds in marine invertebrates from sulfide-rich habitats. *J. Exp. Zool.* 249:121-132.
- Powell, M. A. and G. N. Somero. 1986. Hydrogen sulfide oxidation is coupled to oxidative phosphorylation in mitochondria of *Solemya reidi*. *Science* 233:563-566.
- Reid, R. G. B. and D. G. Brand. 1986. Sulfide-oxidizing symbiosis in Lucinaceans: Implications for bivalve evolution. *Veliger* 29:3-24.
- Schedel, M. and H. G. Truper. 1980. Anaerobic oxidation of thiosulfate and nitrate reductase from *Thiobacillus denitrificans*. *Arch. Microbiol.* 124:205-210.
- Schweimanns, M. and H. Felbeck. 1985. Significance of the occurrence of chemoautotrophic bacterial endosymbionts in lucinid clams from Bermuda. *Mar. Ecol. Prog. Ser.* 24:113-120.
- Somero, G. N., J. J. Childress, and A. E. Anderson. 1989. Transport, metabolism and detoxification of hydrogen sulfide in animals from sulfide-rich marine environments. *Crit. Rev. Aquat. Sci.* 1:591-614.
- Southward, E. C. 1986. Gill symbionts in thyasirids and other bivalve molluscs. *J. Mar. Biol. Ass. U.K.* 66:899-914.
- Southward, E. C. 1987. Contribution of symbiotic chemoautotrophs to the nutrition of benthic invertebrates. In M. A. Sleigh (ed.), *Microbes in the Sea*, Chapter 4. Ellis Horwood, Chichester, England.
- Spiro, B., P. B. Greenwood, A. J. Southward, and P. R. Dando. 1986. $^{13}\text{C}/^{12}\text{C}$ ratios in marine invertebrates from reducing sediments: Confirmation of nutritional importance of chemoautotrophic endosymbiotic bacteria. *Mar. Ecol. Prog. Ser.* 28:233-240.
- Stanley, S. M. 1970. Relation of shell form to life habits of the *Bivalvia* (Mollusca). *Mem. Geol. Soc. Am.* 125:1-293.
- Teal, J. M. and J. W. Kanwisher. 1966. Gas transport in the marsh grass, *Spartina alterniflora*. *J. Exp. Bot.* 17:355-361.
- Turner, R. D. 1985. Notes on molluscs of deep-sea vents and reducing sediments. *Am. Malacol. Bull.* (Special Edition) 1:23-34.
- Vetter, R. D. 1985. Elemental sulfur in the gills of three species of clams containing chemoautotrophic symbiotic bacteria: A possible inorganic energy storage compound. *Mar. Biol.* 88:33-42.
- Vetter, R. D., M. E. Wells, A. L. Kurtzman, and G. N. Somero. 1987. Sulfide detoxification by the hydrothermal vent crab *Bythograea thermydron* and other decapod crustaceans. *Physiol. Zool.* 60:121-137.
- Vetter, R. D., P. A. Matrai, B. Javor, and J. O'Brien. 1989. Reduced sulfur compounds in the marine environment: Analysis by high-performance liquid chromatography. In E. S. Saltzman and W. J. Cooper (eds.), *Biogenic sulfur in the environment*, ACS Symp. Ser. No. 393, pp. 243-261. American Chemical Society, Washington, D.C.